COURSE-BASED PROJECT DOCUMENTATION | DOC-17

Limitations & Future Enhancements

Theoretical Boundaries & Prospective Architectural Improvements

1. Actual Verified Limitations

- ? 1. Syntax-Only Verification: Does not perform semantic analysis, type checking, symbol resolution, or class hierarchy validation.
- ? 2. Single-Error Termination: In accordance with standard pure LALR(1) parsers, parsing halts upon encountering the first syntax error.
- ? 3. No Bytecode Generation: Does not emit JVM .class files or intermediate bytecode.
- ? 4. No Runtime Execution: Does not run, debug, or profile Java programs.
- ? 5. Finalized Java 21 Only: Experimental preview features (e.g. String Templates preview) are excluded.

2. Prospective Future Enhancements (Clearly Labeled as Future Work)

A. Multi-Error Recovery Cascade

Implementing Bison %destructor and error token recovery productions to resynchronize at statement boundaries and report multiple errors in a single pass.

B. Abstract Syntax Tree (AST) Generation & Visualization

Extending grammar action rules to build a hierarchical AST in memory, serializable to JSON, and rendered as an interactive tree in the GUI.

C. Lightweight Semantic Symbol Table Pass

Adding a post-parse symbol table pass to detect undeclared variable references and mismatched argument counts.

D. Cross-Platform Shared Libraries

Extending Makefile targets to build native shared objects (.so for Linux and .dylib for macOS).